



Modeling urban soil physical processes: common challenges and emerging approaches

Sara E. Acevedo^{a,b} & Moreen Willaredt^{c,d}

Organisers:



ISMC 2026

Rio de Janeiro, Brazil

15-19 September 2026

Institutions and Acknowledgements

Sara E. Acevedo^{a,b} & Moreen Willaredt^{c,d}

^aFacultad de Agronomía y Sistemas Naturales, Pontificia Universidad Católica de Chile, 7820436 Santiago, Chile

^bCentro de Desarrollo Urbano Sustentable ANID CEDEUS CIN250009, El Comendador 1916, Providencia, Santiago Chile

^cDepartment of Plant Sciences, University of California, Davis. PES-1238, One Shields Avenue, Davis, CA 95616, USA

^dDepartment of Geography, Humboldt Universität zu Berlin, Berlin, 10117, Germany

Grant: FONDECYT Iniciación 11260726 and VRI-UC support for traveling



Introduction



Context



This presentation focuses on the topics covered in the review article Acevedo et al., 2026

Some examples based on C and soil-water dynamics

REVIEW | MAY 14 2026

Common challenges and potential approaches in urban soil physics research

Collections: Knowledge Domain: Ecology and Earth Systems

Sara E. Acevedo, Beverly Alvarez-Torres, Geoffroy Séré, Moreen Willaredt 

* Email: mwillaredt@ucdavis.edu

Elementa: Science of the Anthropocene (2026) 14 (1): 00099.

<https://doi.org/10.1525/elementa.2025.00099> **Article history** 

<https://doi.org/10.1525/elementa.2025.00099>



Urban soil modelling

Challenges

- Soil degradation
- Lack of data
- Taxonomic inconsistency

Approaches

- Multiscale analysis
 - Trends in determination of urban soil physical properties
- 

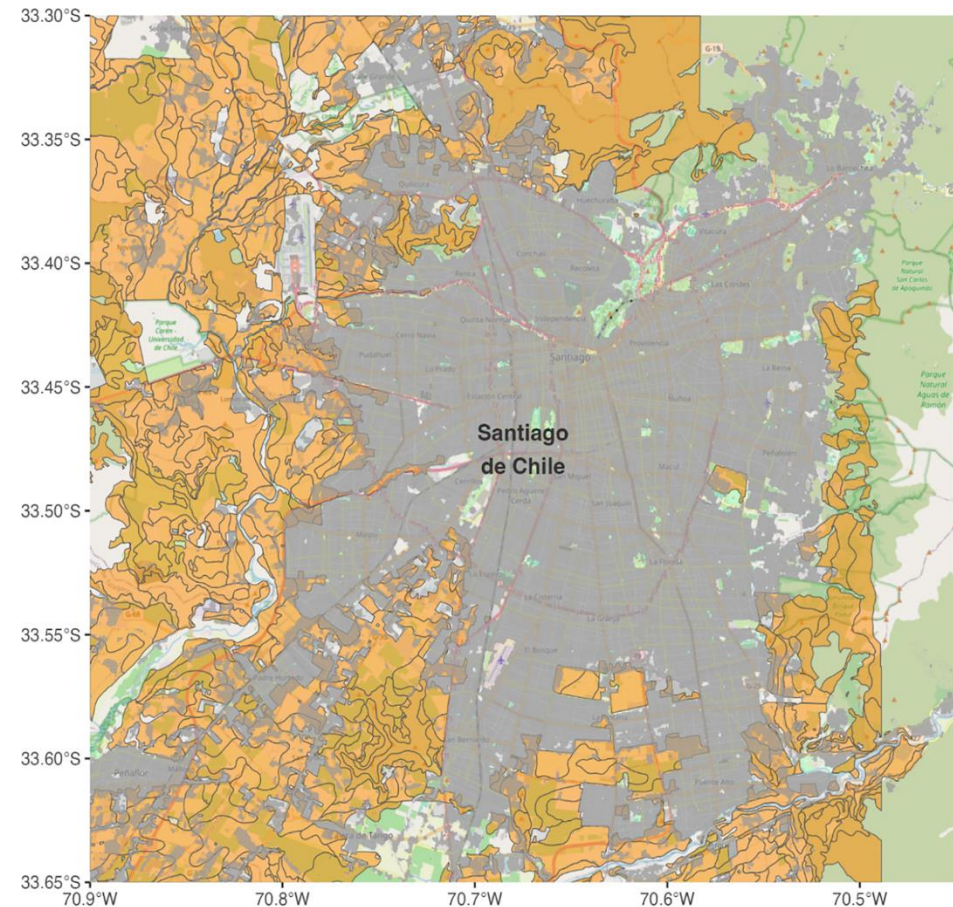
Challenges



Soil degradation and modelling

- Soil sealing is one of the main causes of soil degradation
- Part of urbanization process
- Soil sealing **is not binary**

Map of Santiago de Chile derived from OpenStreetMap. Impervious surface layer derived from the Dynamic World dataset (2020), and Soil Survey layer from Soil Surveys (1990) in the Santiago Metropolitan Area.

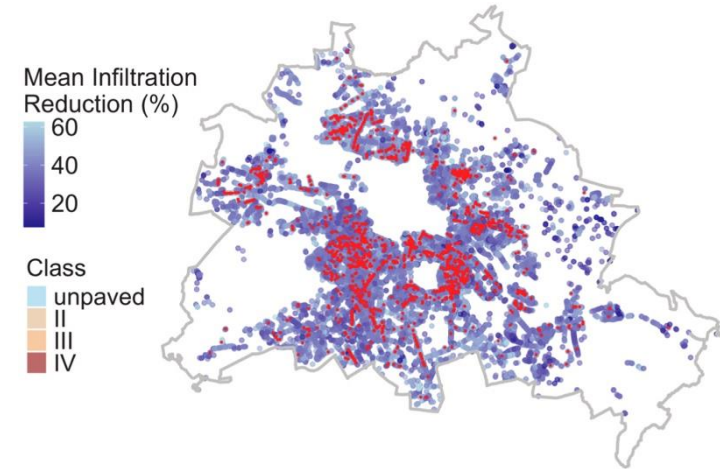
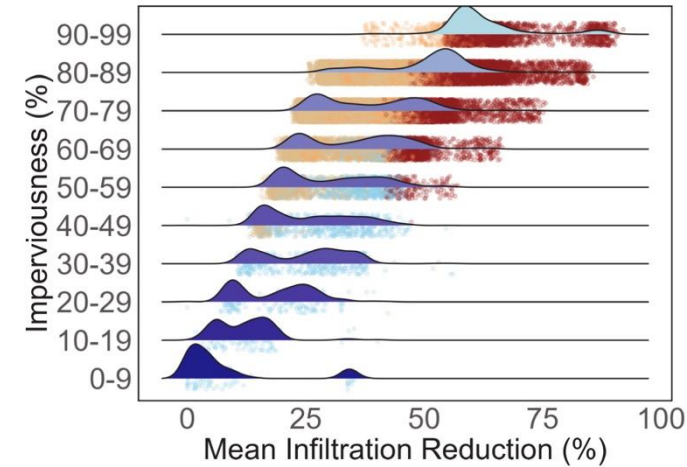


Layer  Impervious surface 2020  Soil Survey 1990

Acevedo et al., 2026

Soil degradation and modelling

- Resolution of globally available satellite derived imperviousness data restricts the differentiation between impervious and partly pervious pavement materials (Acevedo et al., 2026)
- Widely used satellite-derived data underestimates imperviousness in tree catchments (Willaredt et al., 2026)

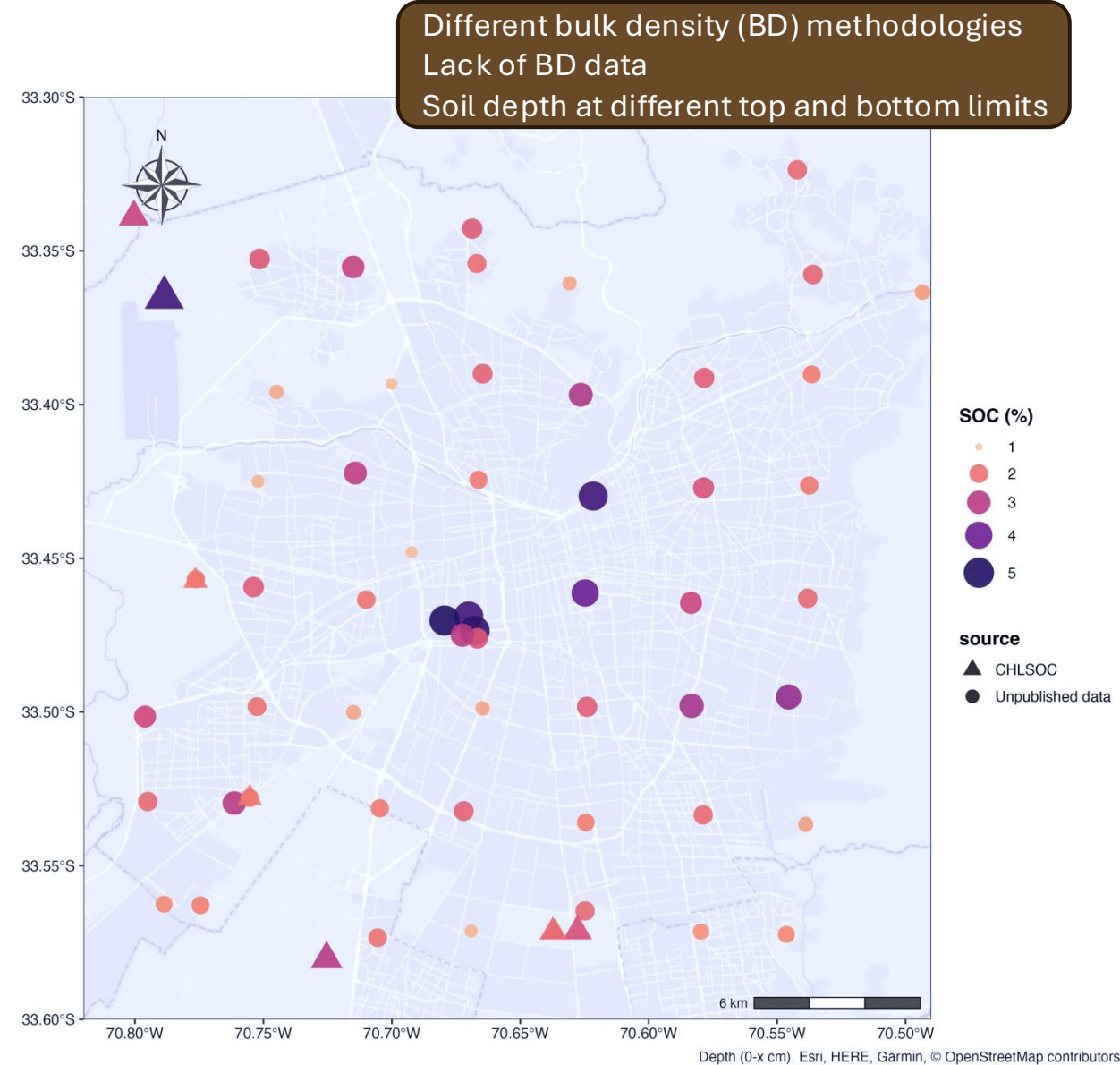


Willaredt et al., 2026

Beyond Imperviousness: spatially explicit assessment of pavement material effects on urban soil water supply for street trees, presentation ISMC 2026

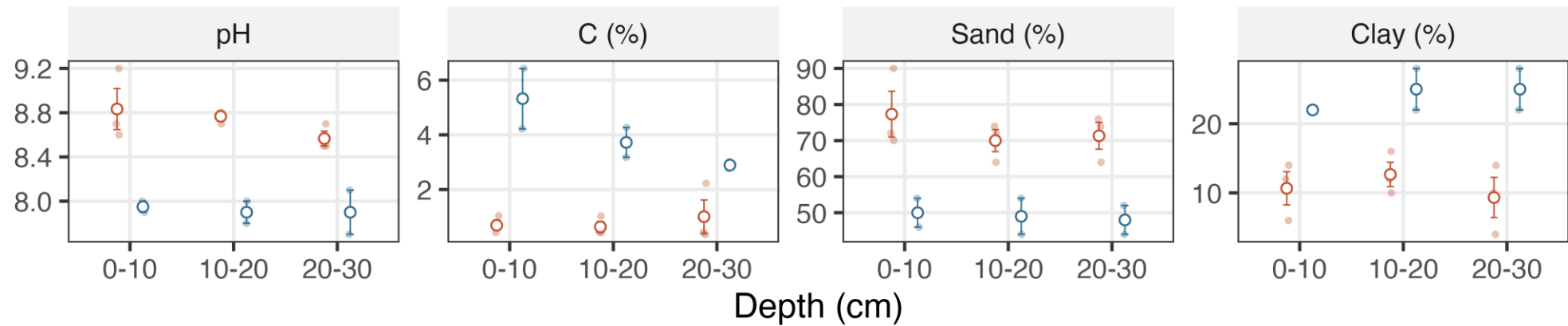
Lack of data

- Different depths
- Harmonization needed for C dynamics
- PTFs not suitable for urban soils or technosols (Jian et al. 2021)



Own elaboration, 2026

Lack of data



Scarce data comparing sealed and unsealed soils in cities

○ PAV ○ Non-PAV

From: Carbon patterns in anthropogenic sealed urban soils: a pilot study, Mariana Dos Santos Moreno, presentation ISMC 2026

Taxonomic inconsistencies

- Urban soils undergo intense pedogenesis and functional approaches like Destisol model (Séré et al., 2024) have shown classical taxonomy fits the urban context poorly.

Soil profile showing abundant boulders of probable alluvial origin, weak differentiation of horizons, and isolated evidence of human activity.



Campus San Joaquín UC Chile, 2026

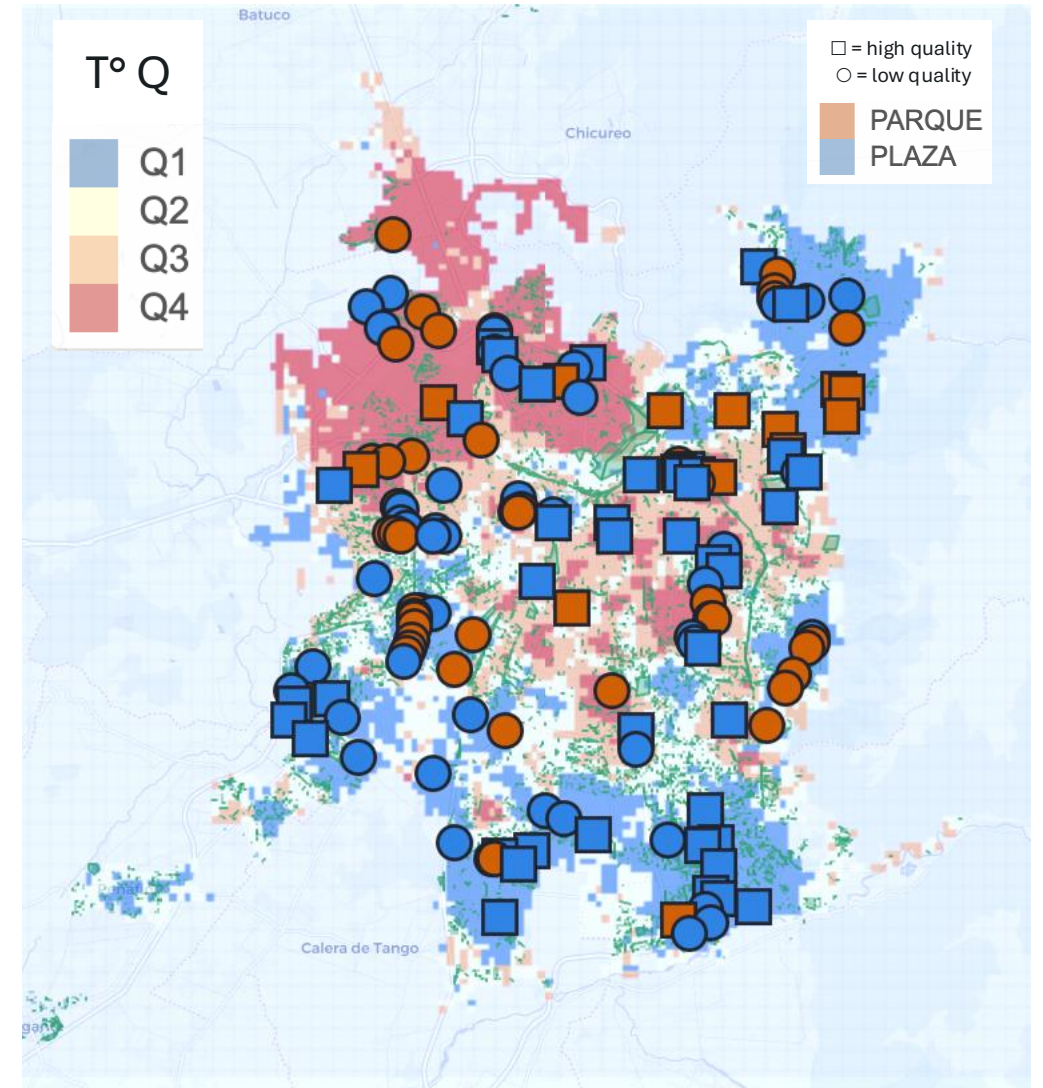
Approaches



Multiscale analysis

To understand the interactions between soil properties, urban ecosystems, and human activities, it is necessary to cover different spatial and temporal scales, from microscopic soil structures to city-wide patterns

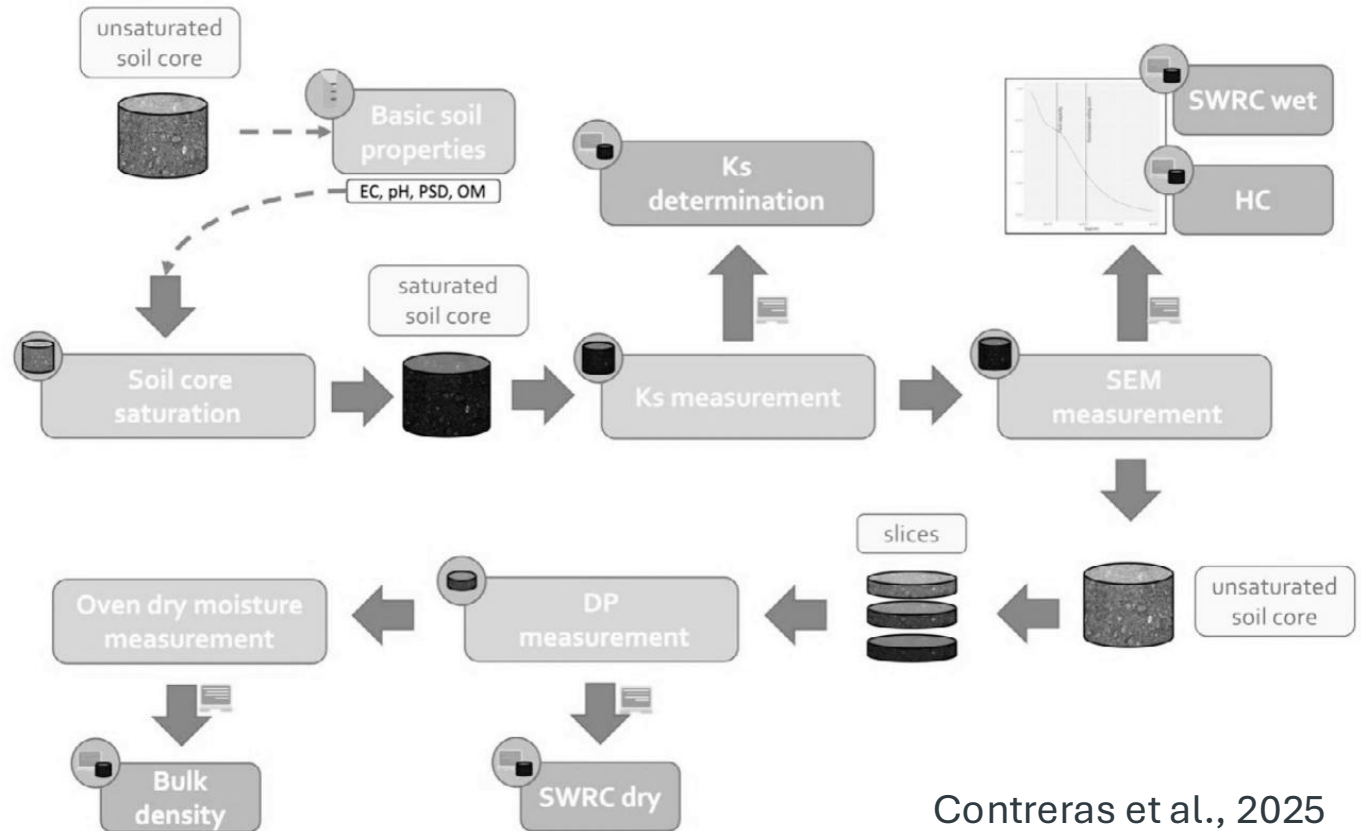
- Urban microclimate
- Socioeconomic (park quality status)



Santiago Chile Park analysis: own elaboration, 2026

Trends in determination of urban soil physical properties

- Presents an in-tandem workflow covering both curves from saturation to dryness
- Aim: reproducibility, lower measurement uncertainty, and standardized practice in soil physics labs, applicable to urban soils



References

Acevedo, S. E., Alvarez-Torres, B., Séré, G., & Willaredt, M. (2026). Common challenges and potential approaches in urban soil physics research. *Elem Sci Anth*, 14(1). <https://doi.org/10.1525/elementa.2025.00099>

Contreras, C. P., Acevedo, S. E., Ávila, C. J., & Bonilla, C. A. (2025). In-tandem methodology and recommendations for measuring soil hydraulic properties from saturation to dryness in laboratory conditions. *International Agrophysics*, 39(2), 203–208. <https://doi.org/10.31545/intagr/200976>

Jian, J., Shiklomanov, A., Shuster, W. D., & Stewart, R. D. (2021). Predicting near-saturated hydraulic conductivity in urban soils. *Journal of Hydrology*, 595, 126051. <https://doi.org/10.1016/j.jhydrol.2021.126051>

Séré, G., Lothode, M., Blanchart, A., Chirol, C., Tribotte, A., & Schwartz, C. (2024). Destisol: A decision-support tool to assess the ecosystem services provided by urban soils for better urban planning. *European Journal of Soil Science*, 75(5). <https://doi.org/10.1111/ejss.13557>

Willaredt, M., Haase, D., & Ossola, A. (2026). High-resolution pavement material data can improve estimates of water supply from precipitation to street trees. *Landscape and Urban Planning*, 276, 105740. <https://doi.org/10.1016/j.landurbplan.2026.105740>



ISMC 2026

5th International Soil Modeling
Consortium Conference



seaceved@uc.cl



Rio de Janeiro, Brazil

Organisers:

